27/3/2026

Assignment:

1] Association Rule Learning:

It is an unsupervised ML technique used to discover relationships or patterns b/w variables in large datasets.

*When to use:

→ Market basket analysis.
→ Recommendation systems.
→ Customer behavior analysis.

* Why to use:

→ Finds hidden patterns.
→ Helps in decision making.

* Working:

→ Identify frequent itemsets.
→ Generate rules.
→ Evaluate rules using metrics.

*Formula:

$X => Y$

where,

X = antecedent.
Y = consequent.

Example:

If a customer buys milk → bread,
Rule: Milk ⟹ Bread.

Scanned with OKEN Scanner

PAGE NO :
DATE :

2] Types of Association Rule Learning :

i] Appriri Algorithm :
→ Uses candidate generation.
→ Removes infrequent items.

ii] ECLAT Algorithm :
→ Uses vertical dataset.
→ Faster than Apriori.

iii] FP-Growth :
→ Uses tree structure.
→ No candidate generation.

3] Evaluation Metrics in Association Rules :

i] Support :
→ Measures how frequently itemset appears.
→ Support (X) = {Transactions containing X} / {Total Transactions}.

ii] Confidence :
→ Probability of Y given X.
→ Confidence $(x \Rightarrow y)$ = Support $(x \cup y)$ / Support (x).

iii] Lift :
→ Measures strength of rule.
→ Lift $(x \Rightarrow y)$ = {Confidence $(x \Rightarrow y)$} / {Support (y)}.

Scanned with OKEN Scanner

PAGE NO :
DATE :

Example:

Transations = 5 in total

Milk appears in 3.

Support = 3/5 = 0.6.

4 Antecedent & Consequent :

* Antecedent :

→ Left side of rule.

→ Condition

Example = Milk

* Consequent :

→ Right side of rule

→ Result.

Example = Bread.

5 Apriori Algorithm :

Apriori is used to find frequent itemsets using minimum support.

* Working Algorithm :

→ Find frequent 1-itemsets

→ Generate candidate 2-itemsets

→ Prune infrequent sets

→ Repeat

* Key Property :

" If subset is not frequent, superset cannot be frequent".

Scanned with OKEN Scanner

PAGE NO :
DATE :

6] **ECLAT Algorithm:**

Equivalence Class Transformation (ECLAT) a depth-first search algorithm used for frequent itemset mining.

* Working:
  - → uses vertical format.
  - → Finds intersection of transactions.

* Advantages:
  - → Faster than Apriori.
  - → Less database scans.

Example:

Milk → {T1, T2}.
Bread → {T1, T3}.
Intersection → {T1}.

7] **FP-Growth Algorithm:**

It is a method to find frequent patterns using a tree structure.

* Working:
  - → Scan dataset.
  - → Build ~~Fixed~~ FP-tree.
  - → Extract frequent patterns.

* Use Case:
  - → Supermarket recommendation.
  - → Online shopping suggestions.

Scanned with OKEN Scanner

PAGE NO :
DATE :

* Advantages:
  - → Faster than Apriori.
  - → No candidate generation.

Example:

Transations : T1 → Milk, Bread.
T2 → Milk, Butter.
T3 → Milk, Bread.

Tree formed with common prefixes.

8] Feature Engineering:

It is the process of creating, selecting & transforming features to improve model performance.

* When to use:
  - → Before training model.
  - → When data is raw.

* Why to use:
  - → Improves accuracy.
  - → Reduces noise.

* Technique:
  - → Feature Scaling
  - → Encoding
  - → Feature Selection
  - → Handling Missing values.

Example:

Age → Young / Adult / Old.
Salary → normalized values.

Scanned with OKEN Scanner